EduGenie: Google Gemini Powered Learning Assistant

Project Description

EduGenie is a lightweight AI-powered educational assistant that simplifies learning through generative AI. Designed for students of all academic levels, EduGenie enables users to:

- Ask questions and receive smart, concise answers
- Understand complex concepts through simplified explanations
- Generate quizzes from topics or text
- Receive personalized learning recommendations
- Summarize large educational passages

Built with **FastAPI** for the backend and a simple **HTML+CSS frontend**, EduGenie leverages lightweight and cloud-based AI models for local efficiency and cloud power. It works well on devices like the Mac M1, making it accessible to a broad range of learners and developers.

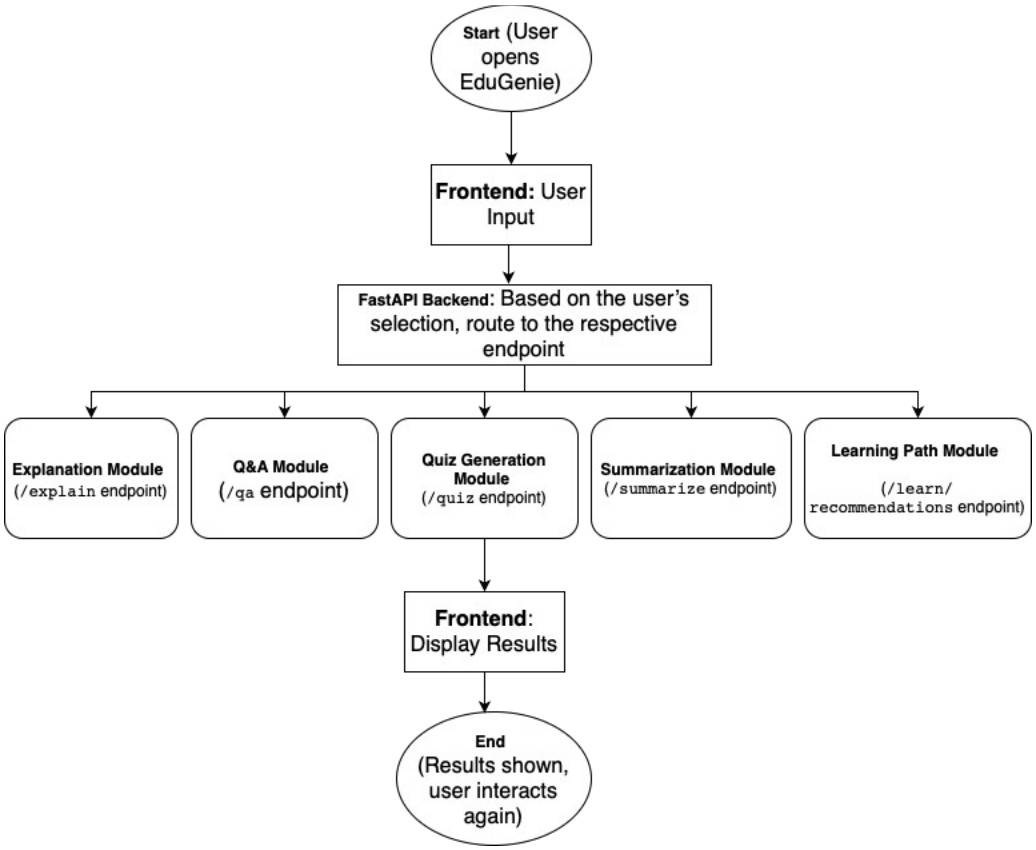
Scenarios

Scenario 1: A student wants to know about oceans and rivers uses EduGenie to ask “Which is the largest ocean?”

Scenario 2: A student wants to know the level of her understanding of “The Pythagoras Theorem” and clicks “Generate Quiz.”

Scenario 3: A learner exploring SQL requests a learning path which is a structured plan with beginner to advanced topics, timelines, and suggestions.

ARCHITECTURAL DIAGRAM



Pre-requisites

1. Python 3.10+
   - Official Documentation: **Google AI for Developers**
   - Installation Guide: **Tom's Hardware**
   - Popular Tutorial: **Python Programming Tutorial - Full Course for Beginners**
   - Installation Steps:

     1. Download the latest Python 3.10+ installer for your operating system from the official website.
     2. Run the installer and ensure you check the box that says "Add Python to PATH".
     3. Follow the installation prompts to complete the setup.
     4. Verify the installation by opening a terminal or command prompt and typing python --version.

2. FastAPI Framework –

- Official Documentation: FastAPI
- User Guide: FastAPI
- Popular Tutorial: FastAPI Crash Course

3. HTML & CSS – Basic templating used in /templates and /static
4. Google Gemini API Key

- Official Documentation: Google AI for Developers
- Setup Guide: GeeksforGeeks
- Popular Tutorial: How to Use Google Gemini API Key
- Setup Steps:
    1. Visit the Google AI Studio and sign in with your Google account.
    2. Create a new project and enable the Gemini API.
    3. Generate an API key and securely store it.

5. Uvicorn (ASGI Server)

- Official Documentation: PyPI
- Installation Steps:
    1. Install Uvicorn using pip: pip install uvicorn

6. Jinja2 (HTML Templating Engine)

- Official Documentation: Jinja2 Documentation
- Installation Steps:
    1. Install Jinja2 using pip: pip install jinja2